

Micro-level Motion Study (Therbligs)

Woojin Park

Life Enhancing Technology Laboratory

SNU Department of Industrial Engineering

What is Motion Study?

- Motion study aims to find the preferred method of doing work
- Focuses on eliminating unnecessary motions
- Improves efficiency and reduces fatigue

Two Levels of Motion Study

- Motion study can be approached at two levels:
 - Micro-level analysis
 - Process-level analysis
- Today's focus: Micro-level analysis

What is Micro-Level Analysis?

- Focuses on human body movements
- Breaks work into smallest motion units
- Useful for repetitive manual tasks
- Enables precise improvement

Micro Motion Study

- Technique for analyzing detailed human motions
- Often uses video or observation
- Based on Therbligs
- Helps identify inefficiencies

Therbligs – Basic Motion Elements

- Developed by Gilbreth
- 17 basic motion elements
- Used to describe all manual work

Effective Therbligs

(Directly advance progress of work. May be shortened but difficult to eliminate completely.)

<u>Therblig</u>	Symbol	Description	Example
Reach	 RE	Motion of empty hand to or from object; time depends on distance moved; usually preceded by Release and followed by Grasp.	
Move	 M	Movement of loaded hand; time depends on distance, weight, and type of move; usually preceded by Grasp and followed by Release or Position.	
Grasp	 G	Closing fingers around an object; begins as the fingers contact the object and ends when control has been gained; depends on type of grasp; usually preceded by Reach and followed by Move.	
Release	 RL	Relinquishing control of object, typically the shortest of the therbligs.	
Preposition	 PP	Positioning object in predetermined location for later use; usually occurs in conjunction with Move, as in orienting a pen for writing.	
Use	 U	Manipulating tool for intended use; easily detected, as it advances the progress of work.	
Assemble	 A	Bringing two mating parts together; usually preceded by Position or Move; followed by Release.	
Disassemble	 DA	Opposite of Assemble, separating mating parts; usually preceded by Grasp and followed by Move or Release.	

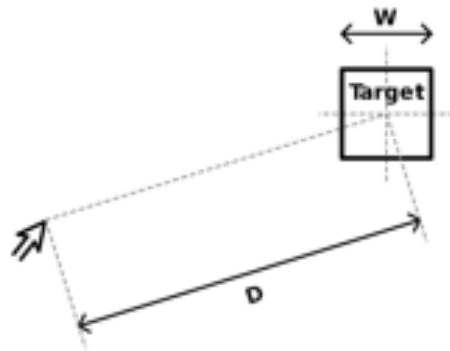
Ineffective Therbligs (Do not advance progress of work. Should be eliminated if possible.)			
<u>Therblig</u>	Symbol	Description	Example
Search	 S	Eyes or hands groping for object; begins as the eyes move in to locate an object.	
Select	 SE	Choosing one item from several; usually follows Search.	
Position	 P	Orienting object during work, usually preceded by Move and followed by Release (as opposed to during for Preposition).	
Inspect	 I	Comparing object with standard, typically with sight, but could also be with the other senses.	
Plan	 PL	Pausing to determine next action; usually detected as a hesitation preceding Motion.	
Unavoidable Delay	 UD	Beyond the operator's control due to the nature of the operation, e.g., left hand waiting while right hand completes a longer Reach.	
Avoidable Delay	 AD	Operator solely responsible for idle time, e.g., coughing.	
Rest to Overcome Fatigue	 R	Appears periodically, not every cycle, depends on the physical workload.	
Hold	 H	One hand supports object while other does useful work.	

Therbligs (Basic Motion Elements)

- **RE (Reach)**

- Also called **Transport Empty (TE)**
- Motion performed with an empty hand
- From the start of movement to arrival/stop
- Example: Reaching for a pen

Fitts' law



$$T = a + b \log_2 \left(2 \frac{D}{w} \right)$$

Annotations for the equation:

- Time* points to T
- Distance* points to D
- Coefficients* points to a and b
- Width* points to w

Therbligs (Basic Motion Elements)

- **G (Grasp)**

- Motion of closing the hand to take control of an object
- From initial contact to full control
- Influenced by:
 - Type of object
 - Shape and weight
- Example: Grasping a pen
- **Improvement ideas:**
 - Contact grasp (without lifting)
 - Use of vacuum, magnets, conveyors
 - Modify surface material (e.g., sponge)
 - Utilize **pre-positioning (PP)**

Therbligs (Basic Motion Elements)

- **M (Move)**

- Moving an object while holding it
- The time required is affected by:
 - Distance
 - Weight
 - Type of movement
- Example: Moving a pen to the paper
- **Improvement ideas:**
 - Mechanization
 - Elimination of unnecessary movement
 - Reduction of distance and weight
 - Elimination of abrupt changes in direction
 - Allowing sufficient tolerance at the stopping point

Therbligs (Basic Motion Elements)

- **RL (Release)**
 - The motion of releasing an object that has been grasped
 - Occurs in a very short moment
 - **Improvement ideas:**
 - Releasing during movement
 - Mechanization

Therbligs (Basic Motion Elements)

- **S (Search)**
 - An eye or hand motion used to locate an object
 - From the beginning of hesitation to the stopping point
 - **Improvement ideas:**
 - Use of color coding
 - Labels
 - Better lighting

Therbligs (Basic Motion Elements)

- **SL (Select)**
 - The motion of choosing one item from several alternatives
 - Sometimes **S (Search)** and **SL (Select)** occur simultaneously
 - Examples:
 - Selecting one pencil from a pencil case
 - Selecting one drill bit from a set of bits
 - **Improvement ideas:**
 - Separate items by type
 - Increase compatibility or standardization

Therbligs (Basic Motion Elements)

- **P (Position)**

- The motion of placing or orienting an object into the correct position
- Includes aligning, turning, or fitting into place
- Examples:
 - Positioning a key or a nail
 - Inserting a subway ticket (in older systems)
- **Improvement ideas:**
 - Eliminate unnecessary positioning
 - Use chamfers (a slanted edge that replaces a sharp corner) or guides to make positioning easier



Therbligs (Basic Motion Elements)

- **PP (Pre-Position)**

- Placing an object in a predetermined location to facilitate the next positioning (P) motion
- Makes subsequent actions faster and easier
- Example:
 - Using a pen holder
- **Improvement ideas:**
 - Use devices such as a **spring tool holder (spring balancer)**
 - Arrange tools in fixed, accessible locations



Therbligs (Basic Motion Elements)

- **A (Assemble)**

- The motion of joining two or more parts together
- From the moment the parts come into contact to the completion of assembly
- Example:
 - Assembling a pen cap with a pen

- **DA (Disassemble)**

- The reverse motion of assembly
- Involves separating previously joined parts

Therbligs (Basic Motion Elements)

- **AD (Avoidable Delay)**

- Delay or interruption that can be controlled by the worker
- Often caused by personal or behavioral factors
- Examples:
 - Coughing
 - Personal habits

Therbligs (Basic Motion Elements)

- **UD (Unavoidable Delay)**

- Delay or interruption beyond the worker's control
- Caused by external factors
- Examples:
 - Delay in material supply
 - Machine waiting time
- **Improvement ideas:**
 - Improve the process or system
 - Address issues at the workflow level rather than the individual level

Therbligs (Basic Motion Elements)

- **U (Use)**
 - The motion of using a tool
 - Involves performing work with equipment
 - **Improvement ideas:**
 - Use jigs and fixtures
 - Use powered tools
 - Improve tool design

Therbligs (Basic Motion Elements)

- **H (Hold)**

- Maintaining an object in position after grasping
- Occurs when one hand holds while the other performs a task
- Example:
 - Holding a bolt while tightening a nut
- **Improvement ideas:**
 - Use tools or fixtures
 - Use both hands effectively
 - Utilize friction or support mechanisms

Therbligs (Basic Motion Elements)

- **PL (Plan)**
 - Mental activity involved in deciding the next action
- **I (Inspect)**
 - Checking or verifying the quality or condition
- **R (Rest)**
 - Recovery from fatigue
 - Necessary for maintaining performance and safety

Productive vs Non-Productive Motions

Classification of Therbligs

Productive Movements

RE (Reach)

G (Grasp)

M (Move)

RL (Release)

PP (Pre-position)

U (Use)

A (Assemble)

DA (Disassemble)

Non-Productive Movements

S (Search)

SE / SL (Select)

P (Position)

I (Inspect)

PL (Plan)

UD (Unavoidable Delay)

AD (Avoidable Delay)

R (Rest)

H (Hold)

*Goal: eliminate or reduce waste

Principles of Motion Economy

- Minimize the **number and distance of motions**
- Use **both hands simultaneously** and symmetrically
- Perform finger motions **from outside toward the body** for better accuracy and strength
- Use **smooth and continuous motions**
- Design motions that can be **easily learned and habitual**
- Minimize **control motions** (e.g., carrying a cup of coffee carefully)
- Reduce **eye fatigue**
- Use **different parts of the body alternately** to avoid fatigue

Example – Task Improvement

Example – Task Improvement Using Therbligs

- Task: Charging a smartphone
- Objective: Analyze and improve the task using Therbligs
- Focus:
 - Identify non-productive motions
 - Redesign the task to improve efficiency

Task Scenario (Before Improvement)

- A user plugs in a smartphone to charge
- Charging cable is placed on the desk
- Multiple cables may be present
- Connector requires precise alignment

Therblig Analysis (Before)

- S (Search): Look for charging cable
- SE (Select): Choose correct cable
- RE (Reach): Reach for cable
- G (Grasp): Pick up cable
- M (Move): Move cable toward phone
- P (Position): Align connector
- I (Inspect): Check orientation
- Repeat P + I if misaligned
- RL (Release): Release cable

Problems Identified

- Presence of non-productive motions:
 - Search (S), Select (SE)
 - Position (P), Inspect (I)
- Repeated alignment attempts
- High precision required
- Increased cognitive load

Improvement Idea

- Introduce magnetic charging system (e.g., MagSafe)
- Use fixed cable location
- Enable automatic alignment



Therblig Analysis (After Improvement)

- RE (Reach)
- G (Grasp)
- M (Move)
- RL (Release)

*Position and Inspect are eliminated

Before vs After Comparison

Before

Search, Select required

Multiple Position + Inspect

Precise alignment needed

High cognitive effort

After

Eliminated

Eliminated

Automatic alignment

Minimal effort

Key Insight

- Good design eliminates non-productive motions
- Focus on:
 - Removing unnecessary steps
 - Simplifying interaction
- Design can replace human effort

Design Principles Applied

- Eliminate Search → fixed location
- Eliminate Select → standardization
- Reduce Position → magnetic guidance
- Reduce Inspect → automatic feedback

Takeaway

- Therbligs provide a powerful tool for task analysis
- Even simple daily tasks can be improved
- Motion study is highly relevant to modern product design

Key Takeaway

- Micro-level analysis focuses on human motion
- Therbligs provide a structured framework
- Foundation for improving manual work

Comparing Paper- and AR-Based Assembly Manuals in Task Performance, dIPFC Hemodynamic Responses, and Perceived Workload

Human Factors
2026, Vol. 0(0) 1–21
© 2026 Human Factors
and Ergonomics Society
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/00187208261439486
journals.sagepub.com/home/hfs



Jungmin Ryu¹ , Philjun Moon¹, Kitae Hwang¹ , Sojeong Yoon¹,
Yoonjin Lee¹ , Qinghua Piao¹, Jin-Hyuck Park², and Woojin Park^{1,3}

Abstract

Objective: This study comparatively evaluated three assembly manual types (traditional paper-based, user-fixed AR, and world-fixed AR) in terms of task performance, dIPFC hemodynamic responses, and perceived workload.

Background: Prior studies of AR and traditional assembly manuals have relied mainly on subjective workload ratings, providing only an indirect view of cognitive effort. Neurophysiological assessment can quantify mental workload more directly. Measuring dIPFC activation may offer a process-specific index of cognitive control demands.

Method: Twenty-four participants performed a 7-min modified Purdue Pegboard Test with each manual type in a within-subjects design. Task performance, dIPFC hemodynamic responses, and perceived workload were assessed. fNIRS was used to measure hemodynamic responses.

Results: The world-fixed AR manual yielded significantly better task performance than both paper-based and user-fixed AR manuals, particularly in reducing location errors. Both AR manuals elicited significantly higher dIPFC activation than the paper-based manual, indicating elevated cognitive control demands. Perceived workload did not differ significantly across manual types.

Conclusion: The world-fixed AR manual improved task performance by reducing information access cost, yet both AR manuals increased cognitive control demands relative to the paper-based manual, likely due to perceptual conflicts inherent in AR. The user-fixed AR manual did not outperform the paper-based version, indicating that AR effectiveness depends on design specifics.

Application: The study recommends the world-fixed AR manual among the three types examined while emphasizing the need for innovations that reduce perceptual conflicts. It also shows that neurophysiological assessment provides insights beyond task performance and subjective ratings, supporting multidimensional evaluation to optimize interface technologies.

¹Seoul National University, South Korea

²Soonchunhyang University, South Korea